



Segmented fast linear canonical transform

YAN-NAN SUN^{1,2} AND BING-ZHAO LI^{1,2,*}¹School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China²Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China

*Corresponding author: li_bingzhao@bit.edu.cn

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Investigation of the discrete and fast linear canonical transforms is becoming one of the hottest research topics in modern signal processing and optics. How to handle and obtain the linear canonical frequency spectrum of very large input data based on equipment with limited memory space is one of the key problems. To focus on this problem, a new kind of segmented fast linear canonical transform has been proposed in this paper. First, the large data is segmented into short data. Thereby, the proposed algorithms can calculate very large input data and simultaneously keep the ideal frequency resolution. Second, the complexity of the derived algorithms has been analyzed in detail for different kinds of signals. Their performance with regard to resolution and precision are compared with the existing fast linear canonical transforms. Finally, experimental results are presented to verify the correctness of the results obtained. © 2018 Optical Society of America

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1. INTRODUCTION

The linear canonical transform (LCT), which describes the effect of quadratic phase systems on a wave field, is a parameterized general linear integral transform with three degrees of freedom [1–3]. In optics, the LCT can describe a broad class of optical systems, including thin lenses, sections of free space in the Fresnel approximation, sections of quadratic graded-index media, and linear quadratic phase systems [3–5]. The LCT generalizes many optical transforms, such as Fourier and fractional Fourier transforms, coordinate scaling, chirp multiplication, and Fresnel transform [6,7]. These integral transforms are of great importance in electromagnetic, acoustic, and other wave propagation problems, where they represent the solution of the wave equation under a variety of circumstances. The LCT itself has also found many applications in the solution of optical systems, radar system analysis, filter design, pattern recognition, image watermarking, and many others fields [8,9]. Basic theories of LCT have been developed that include convolution theorems [10–12], sampling theorems [6,13–17], and uncertainty principles [18,19]. The numerical approximation of the LCT is of importance in modeling first-order optical systems and many signal processing applications.

Therefore, since the continuous LCT was introduced, the definition and implementation of the discrete linear canonical transform (DLCT) have been widely considered by many researchers. The existing discrete algorithms are divided mainly into the following categories. The first one is the direct discretization algorithm [20–22], which converts directly integral

into summation. The second one decomposes an arbitrary LCT operator into its special cases that have a fast algorithm [23–26]. The third one is a DLCT-based algorithm [27–30], which is similar to employing the fast Fourier transform (FFT) to implement the discrete Fourier transform. In addition, some other algorithms have also been proposed in recent years [31–38]. To the best of our knowledge, all existing algorithms can effectively and rapidly calculate the entire spectrum in the LCT domain. However, two main practical issues associated with the discrete and fast LCT still remain unresolved. The first one is that these algorithms are usually limited by the storage size of the analysis equipment. Generally speaking, the theoretical LCT frequency resolution is proportional to sample points N . If the device allows the algorithm to handle only $M < N$ points at one time, theoretical LCT frequency resolution can no longer be realized, so some information will be lost. The other issue is how to fast calculate the **local** LCT spectrum according to the practical requirement. In practical applications, we sometimes need to observe the local LCT spectrum in detail.

To overcome the aforementioned problems, we propose a kind of segmented fast linear canonical transform (SFLCT), which converts the single DLCT of a long sequence into multiple pairs of DLCT of short sequence. In this processing, each pair of short sequences of the DLCT corresponds to a LCT frequency cell. Therefore, the SFLCT has the ability to limit its calculation to a portion of the frequency spectrum of interest in the LCT domain. Since only a small size of DLCT is required, the storage space will be substantially reduced.

Thereby, no matter how large is the input data size, only very limited memory space is required. To further improve the implementation and computation complexity of the SFLCT, we derive the zoom fast linear canonical transform (ZFLCT), which is based on the idea of segmentation for a specific signal. Owing to using all input data for calculating each LCT frequency cell, the SFLCT and ZFLCT have ideal frequency resolution.

The rest of the paper is organized as follows. In Section 2, a brief review of the related preliminaries is presented. In Section 3, for different kinds of signal, the mathematical formulas of the SFLCT and ZFLCT are derived by different segmentation methods, and the computational complexity is also analyzed in detail. Their performance with regard to resolution and precision are compared with the existing fast linear canonical transforms. In Section 4, the simulations are presented. Finally, conclusions are drawn in Section 5.

2. PRELIMINARIES

A. Linear Canonical Transform

In this paper, the LCT of a signal $x(t)$ with three parameters α , β , and γ , is defined as [3]

$$X^A[x(t)](u) = \int_{-\infty}^{\infty} x(t) K_A(u, t) dt,$$

where $K_A(u, t) = \sqrt{\beta} e^{-j\frac{\pi}{4}} e^{j\pi(\alpha u^2 - 2\beta ut + \gamma t^2)}$ is the kernel of the LCT. The parameters α , β , and γ are real and independent of t and u . $A = [\gamma/\beta, 1/\beta; -\beta + \alpha\gamma/\beta, \alpha/\beta]$ is a 2×2 real matrix, and $\det(A) = 1$; in such case, the LCT operator is unitary in $L^2(R)$, and $K_{A^{-1}}(t, u) = K_A^*(u, t)$, where the $K_A^*(u, t)$ is the conjugation transpose of $K_A(u, t)$, and A^{-1} is the inverse of A . Thus, the inverse LCT is given by the LCT with parameter matrix A^{-1} :

$$x(t) = \int_{-\infty}^{\infty} X^A[x(t)](u) K_{A^{-1}}(t, u) du. \quad (1)$$

When $\alpha = \gamma = 0$, $\beta = 1$, the LCT reduces to the Fourier transform, and converts into the fractional Fourier transform when $\alpha = \gamma = \cot \theta$, $\beta = \csc \theta$. The LCT also reduces to the Fresnel transform if $\alpha = \beta = \gamma = 1/\lambda z$. One can refer to Refs. [7,39] for a detailed description of the relations between the LCT and its special cases.

B. Discrete Linear Canonical Transform

The LCT of the ideal uniform sampled signal with sampling interval Δt is given by [27,29]

$$\begin{aligned} \tilde{X}^A[x(n\Delta t)](u) &= X^A \left[\sum_{n=-\infty}^{\infty} x(n\Delta t) \delta(t - n\Delta t) \right] (u) \\ &= \sum_{n=-\infty}^{\infty} K_A(u, n\Delta t) x(n\Delta t) \\ &= \frac{1}{\Delta t} e^{j\pi\alpha u^2} \sum_{n=-\infty}^{\infty} X^A \left(u - \frac{n}{\Delta t\beta} \right) e^{-j\pi\gamma \left(u - \frac{n}{\Delta t\beta} \right)^2}, \end{aligned} \quad (2)$$

where

$$K_A(u, n\Delta t) = e^{j\pi\alpha u^2} e^{-j2\pi\beta u n\Delta t} e^{j\pi\gamma (n\Delta t)^2}.$$

Equation (2) shows that the LCT of the uniform sampled signal replicates $X^A(u)$ with a period of $1/(\Delta t\beta)$. By sampling uniformly N points in the replicated period $1/(\Delta t\beta)$ in the LCT domain, the sampling interval Δu in the LCT domain is $1/(\Delta tN|\beta|)$. The DLCT of $x(n\Delta t)$ is defined as [20,21,24,27,32]

$$X^A(m) = \sum_{n=0}^{N-1} K_A(m, n) x(n), \quad (3)$$

where $m = 0, 1, 2, \dots, N-1$,

$$X^A(m) = X^A[x(n\Delta t)](m\Delta u), \quad x(n) = x(n\Delta t),$$

$$K_A(m, n) = K_A(m\Delta u, n\Delta t) = e^{j\pi[\alpha(m\Delta u)^2 - 2\beta mn\Delta t\Delta u + \gamma(n\Delta t)^2]}.$$

If we take $\alpha = \alpha \frac{\Delta u}{\Delta t}$, $\beta = \beta$, $\gamma = \gamma \frac{\Delta t}{\Delta u}$, the discrete transform kernel $K_A(m\Delta u, n\Delta t)$ can be simplified as

$$K_A^N(m, n) = e^{j\frac{\pi}{N|\beta|}(\alpha m^2 - 2\beta nm + \gamma n^2)}.$$

According to the sampling theorem [22,40,41], $f = 1/\Delta t$ is often called the Nyquist rate associated with the LCT, and $\Delta u = 1/(N|\beta|\Delta t)$ is the frequency resolution in the LCT domain. This DLCT can be fast computed by performing a chirp multiplication, a FFT, and a second chirp multiplication (CM), denoted FLCT [24]. The computational burden is $2N + N/2 \log_2 N$.

3. MAIN RESULTS

In this section, the SFLCT and the ZFLCT are established by different segmentation methods. The implementation and complexity of the proposed algorithms are discussed in detail. It is noted that the so-called twiddle factor [42] and one of the exponential factors associated with LCT transform parameters are treated in a completely different manner for the two algorithms proposed.

A. Segmented Fast Linear Canonical Transform

In this subsection, the SFLCT is developed for arbitrary kinds of signals, which requires rearranging the order of input data.

1. Derivation of SFLCT

The DLCT of any discrete signal $x(n)$ is defined as in Eq. (3). Now, if sampling points N is a composite number, with factors p and q , it is possible to decompose the input sample points into p blocks of q points as follows:

$$\begin{aligned} n &= sp + v, \quad s = 0, 1, 2, \dots, q-1, \\ v &= 0, 1, 2, \dots, p-1. \end{aligned} \quad (4)$$

Thus, Eq. (3) is rewritten as

$$X^A(m) = \sum_{v=0}^{p-1} \sum_{s=0}^{q-1} K_A(m, sp + v) x(sp + v). \quad (5)$$

Based on the relationship between Δt and Δu , the discrete transform kernel $K_A(m, sp + v)$ can be reformulated as

$$\begin{aligned} K_A(m, sp + v) &= K_A^N(m, sp + v) \\ &= e^{j\pi\alpha' \frac{m^2}{N|\beta|}} e^{j\pi\gamma' \frac{(sp+v)^2}{N|\beta|}} e^{-j2\pi \frac{m(sp+v)}{N}}, \end{aligned} \quad (6)$$

where N is the length of the discrete data $x(n)$,

$$\alpha' = \alpha \frac{\Delta u}{\Delta t}, \quad \gamma' = \gamma \frac{\Delta t}{\Delta u},$$

$$\hat{A} = [\gamma'/\beta, 1/\beta; -\beta + \alpha'\gamma'/\beta, \alpha'/\beta]. \quad (7)$$

Equation (5) can be factorized as

$$X^A(m) = \sum_{v=0}^{p-1} \sum_{s=0}^{q-1} K_A^N(m, sp+v)x(sp+v). \quad (8)$$

To utilize the FLCT to calculate the inner layer summation of Eq. (8), the LCT frequency index m is transformed as

$$m = aq + b; \quad a = 0, 1, 2, \dots, p-1;$$

$$b = 0, 1, 2, \dots, q-1. \quad (9)$$

Thus, Eq. (8) is rewritten by

$$X^A(aq+b) = e^{j\pi\alpha'\frac{2ab}{p|\beta|}} \sum_{v=0}^{p-1} e^{j\pi\left(\frac{\alpha'(aq)^2}{N|\beta|} - \frac{2av}{p} + \frac{\gamma'^2}{N|\beta|}\right)} e^{-j2\pi\frac{bv}{N}}$$

$$\times \sum_{s=0}^{q-1} x(sp+v) e^{j\pi\gamma'\frac{2sv}{q|\beta|}} e^{j\pi\left(\frac{\gamma'(sp)^2}{N|\beta|} - \frac{2sb}{q} + \frac{\alpha'b^2}{N|\beta|}\right)}. \quad (10)$$

This is the basic equation from the DLCT algorithm in Refs. [24,27]. We can further reformulate Eq. (10) as

$$X^A(aq+b) = e^{j\pi\alpha'\frac{2ab}{p|\beta|}} \sum_{v=0}^{p-1} Y(v,b) K_{A''}^p(a,v) e^{-j2\pi\frac{bv}{N}}, \quad (11)$$

where

$$Y(v,b) = \sum_{s=0}^{q-1} x(sp+v) e^{j\pi\gamma'\frac{2sv}{q|\beta|}} K_{A'}^q(b,s), \quad (12)$$

$$K_{A'}^q(b,s) = e^{j\pi\alpha'\frac{b^2}{N|\beta|}} e^{j\pi\gamma'\frac{(sp)^2}{N|\beta|}} e^{-j2\pi\frac{bs}{q}},$$

$$K_{A''}^p(a,v) = e^{j\pi\alpha'\frac{(aq)^2}{N|\beta|}} e^{j\pi\gamma'\frac{v^2}{N|\beta|}} e^{-j2\pi\frac{av}{p}},$$

$$A' = [p\gamma'/\beta, 1/\beta; -\beta + \alpha'\gamma'/\beta, \alpha'/\beta p],$$

$$A'' = [\gamma'/q\alpha', 1/\beta; -\beta + \alpha'\gamma'/\beta, \alpha'q/\beta].$$

In this case, the twiddle factor [42] is contained in the second summation. Meanwhile, we note that the first summation is performed on the index s in Eq. (10); thus, the sample points $x(v), x(p+v), x(2p+v)$, etc., are first summed, which requires rearranging the input data.

For the sake of brevity, we can reformulate Eqs. (11) and (12) as the form of a transformation matrix:

$$\mathbf{Y}_v(b) = [Y(v,0), Y(v,1), \dots, Y(v,q-1)]^T = \mathbf{V}\mathbf{x}_v(s), \quad (13)$$

$$\tilde{\mathbf{Y}}_v(b) = \left[Y(v,0), Y(v,1)e^{-j2\pi\frac{v}{N}}, \dots, Y(v,q-1)e^{-j2\pi\frac{(q-1)v}{N}} \right]^T, \quad (14)$$

$$\mathbf{X}_b^A(a) = \mathbf{D}^b \mathbf{U} \tilde{\mathbf{Y}}_b(v), \quad (15)$$

where

$$\tilde{\mathbf{Y}}_b(v) = \left[Y(0,b), Y(1,b)e^{-j2\pi\frac{b}{N}}, \dots, Y(p-1,b)e^{-j2\pi\frac{(p-1)b}{N}} \right]^T,$$

$$\mathbf{x}_v(s) = \left[x(v), x(p+v)e^{j\pi\gamma'\frac{2v}{q|\beta|}}, \dots, x(N-p+v)e^{j\pi\gamma'\frac{2(q-1)v}{q|\beta|}} \right]^T,$$

$$\mathbf{X}_b^A(a) = [X^A(b), X^A(q+b), \dots, X^A((p-1)q+b)]^T, \quad (16)$$

$\{\mathbf{D}^b\}_{b=0}^{q-1}$ are $p \times p$ diagonal matrices, and

$$[\mathbf{D}^b]_{a,a} = e^{-j2\pi\alpha'\frac{ab}{p|\beta|}}, \quad a = 0, 1, 2, \dots, p-1,$$

$$b = 0, 1, \dots, q-1.$$

The matrices $\mathbf{U}$ and $\mathbf{V}$ are, respectively, the kernel matrix of p -point DLCT with parameters $\alpha'q, \beta$, and γ'/q , and that of q -point DLCT with parameters $\alpha'/p, \beta$, and $p\gamma'$. Equations (13) and (15) constitute multiple pairs of DLCT, and these DLCT can implement quickly using FLCT. Equation (15) associated with Eqs. (13) and (14) is called SFLCT. Since it needs to calculate only the small size of DLCT according to the practical requirements, this algorithm can substantially reduce the need for storage space.

2. Implementation and Complexity Analysis of SFLCT

From Eqs. (11) and (12), we note that $x(sp+v)$ is seen as a two-dimensional representation of input sequence $x(n)$, and the corresponding matrix form is denoted as $\mathbf{x}_{q \times p}$; the s th row and v th column element of $\mathbf{x}_{q \times p}$ is $x(sp+v)$. We calculate the entire LCT spectrum by the following steps:

Step 1: Each element of matrix $\mathbf{x}_{q \times p}$ multiplied by the $e^{j\pi\gamma'2sv/(q|\beta|)}$, denoted by

$$\tilde{\mathbf{x}}_{q \times p} = \{x(sp+v)e^{j\pi\gamma'2sv/(q|\beta|)}\}_{s=0, v=0}^{q-1, p-1}.$$

Step 2: Perform q -point FLCTs of all columns of $\tilde{\mathbf{x}}_{q \times p}$ with parameters $\alpha'/p, \beta, p\gamma'$, denoted by

$$\mathbf{Y} = [\mathbf{Y}_0(b), \mathbf{Y}_1(b), \mathbf{Y}_2(b), \dots, \mathbf{Y}_{p-1}(b)].$$

Step 3: Each element of $\mathbf{Y}$ in Step 2 is multiplied by the phase rotation factor $e^{-j2\pi bv/N}$, $b = 0, 1, 2, \dots, q-1$, $v = 0, 1, 2, \dots, p-1$, denoted by

$$\tilde{\mathbf{Y}} = [\tilde{\mathbf{Y}}_0(b), \tilde{\mathbf{Y}}_1(b), \dots, \tilde{\mathbf{Y}}_{p-1}(b)].$$

Step 4: Perform FLCT for every column of transpose $\tilde{\mathbf{Y}}$ with parameters $\alpha'q, \beta, \gamma'/q$. The results are modulated by $e^{j\pi\alpha'2ab/(p|\beta|)}$, and denoted as

$$\mathbf{X}_{p \times q}^A = [\mathbf{X}_0^A(a), \mathbf{X}_1^A(a), \dots, \mathbf{X}_{q-1}^A(a)]^T,$$

where $\mathbf{X}_b^A(a)$, $b = 0, 1, 2, \dots, q-1$ is Eq. (16). All row elements of matrix $\mathbf{X}_{p \times q}^A$ constitute the entire LCT spectrum.

For calculating the entire LCT spectrum, the processing needs some modulation and FLCT operation. The corresponding total computational complexity is $7N + N/2 \log_2(N)$. It is noted that if the storage size of the analysis equipment is large enough, then the FLCT has less calculational burden than SFLCT for calculating the entire spectrum. However, if the length of an input sequence N exceeds the storage capacity of the analytical devices, then the SFLCT is more effective. Based on these facts, we conclude that the SFLCT is a good choice for spectral analysis of large data.

In addition, we can select specific a_0 , while b varies from 0 to $q - 1$ in Eq. (15) to obtain a subset of the LCT spectrum. Thus, we can utilize a method different than the above to implement the SFLCT. The implementation keeps Step 3 of the previous method, while the following two steps are required:

Step 4: Perform the DLCT with parameters $\alpha'q$, β , and γ'/q for the a_0 th row of the matrix $\tilde{\mathbf{Y}}$ by the directly weighted summation in Eq. (11), denoted by $\mathbf{X}_{a_0}^A(b)$, $b = 0, 1, \dots, q - 1$.

Step 5: Multiply by phase rotation factor $e^{j\pi\alpha'2a_0b/(p|\beta|)}$ every element of the result in Step 4, denoted as

$$\mathbf{X}_{a_0}^A(b) = [X^A(a_0q), X^A(a_0q + 1), \dots, X^A(a_0q + q - 1)]^T. \quad (17)$$

Thus, the a_0 th frequency cell is obtained.

The SFLCT can implement selectivity of spectrum output by selecting different a . It is noted that if all input samples are used to obtain the LCT spectrum in a specific frequency band, the ideal resolution $f/N|\beta|$ can be reached. If the length of an input sequence N exceeds the storage capacity of the analytical devices, the SFLCT is effective, whether it is the calculation of the local or entire spectrum.

For fast calculation of the local LCT spectrum, the processing needs some phase modulation and FLCT operation. If T_i and $M(i)$ are, respectively, the number of computations required to do an FLCT and modulation operation of i points, the total computational requirement of SFLCT is calculated as

$$\begin{aligned} T &= qT_p + 2qM(p) + M(q) + pq \\ &= 5N + N/2 \log_2(q) + q. \end{aligned}$$

To produce the same resolution, if the core storage can handle N sample points, we require direct N -point FLCT, and the corresponding the amount of calculations T_N is

$$T_N = 2N + \frac{N}{2} \log_2 p + \frac{N}{2} \log_2(q),$$

and the calculation savings is given by

$$T_N - T = \frac{N}{2} \log_2(p) - q - 3N.$$

For fixed N , we change p and q to control the length of the spectrum cell.

When $N = 20480$, the numbers of calculations of SFLCT and FLCT are shown in Fig. 1(a) for different lengths of frequency cell q , and the computational savings of SFLCT is plotted in Fig. 1(b). The results suggest that the fewer points we need to calculate, the better the SFLCT is.

B. Zoom Fast Linear Canonical Transform (ZFLCT)

To improve the implementation and the complexity of SFLCT, we propose the ZFLCT, which does not require rearranging the order of input data.

1. Proposed ZFLCT

In this subsection, the data $x(n)$ is split into q blocks of p points. The indices n and m still are transformed as in Eqs. (4) and (9), but the order of summation is different from SFLCT. Thus, the original transform, Eq. (3), is reformulated as

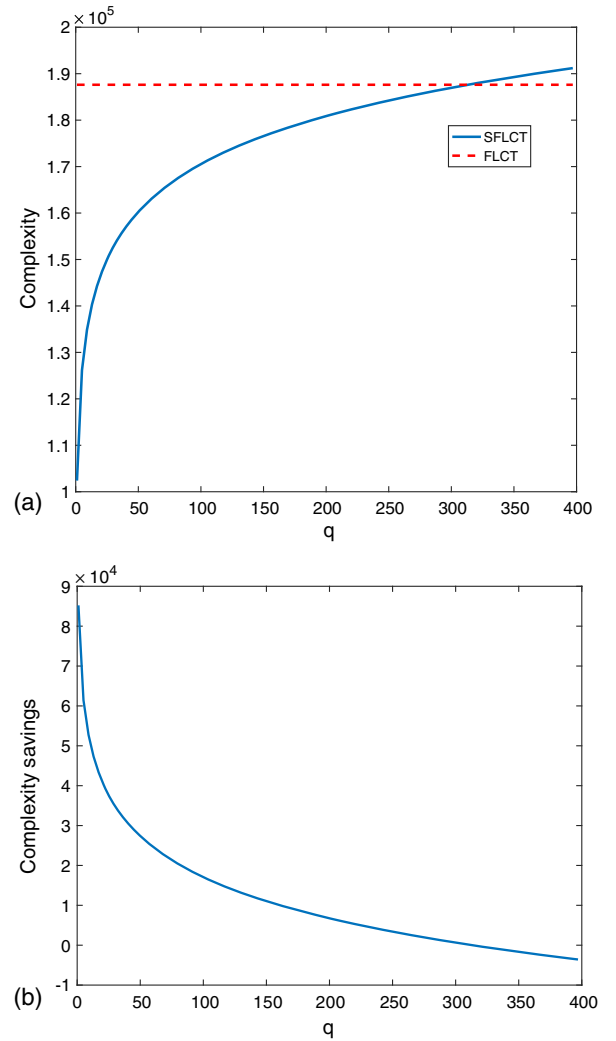


Fig. 1. When $N = 20480$ and $q = 1, 2, 4, \dots, 400$, (a) the number of calculations of FLCT and SFLCT and (b) computational savings between FLCT and SFLCT.

$$\begin{aligned} X^A(aq + b) &= e^{j\pi\alpha' \frac{2ab}{p|\beta|}} \sum_{s=0}^{q-1} K_{A'}^q(b, s) \\ &\times \sum_{v=0}^{p-1} K_{A'}^p(a, v) e^{-j2\pi \frac{av}{N}} x(sp + v) e^{j\pi\gamma' \frac{2av}{q|\beta|}}. \end{aligned} \quad (18)$$

From Eq. (18), the twiddle factor $e^{-j2\pi vb/N}$ is contained in the first summation over v . The storage requirement for the evaluation of Eq. (18) can be estimated roughly. On the summation over v , a and b represent indices of frequency components to cover all components $\max(a) = p - 1$, $\max(b) = q - 1$. Therefore, the storage requirement for output of the v summation, if we need all frequency components, is $pq = N$ points, which is equivalent to the storage requirement for direct N -point FLCT. This is not available if the N input points already had to be sectioned.

Thereby, we take $e^{-j2\pi vb/N} = 1$ and $e^{j\pi\alpha'2aqb/(N|\beta|)} = 1$ in Eq. (18) to make the summation over v independent of b . This effectively cuts the storage requirement down by p . Equation (18) can be rewritten as

$$Y_1(a, s) = \sum_{v=0}^{p-1} x'(sp + v) K_{A''}^p(a, v), \quad (19)$$

$$Z^A(aq + b) = \sum_{s=0}^{q-1} K_{A'}^q(b, s) Y_1(a, s), \quad (20)$$

where $x'(sp + v) = x(sp + v) e^{j\pi\gamma'2sv/(q|\beta|)}$.

Equation (20) is called ZFLCT. It is noted that the sample points enter the summation in chronological order, and shuffling of data points is not necessary. Because of reducing the operation, the ZFLCT has better performance in terms of implementation and computation burden than SFLCT.

In practice, a is the component of the LCT frequency index of the s th p -point DLCT with parameters $\alpha'q, \beta, \gamma'/q$. There are q such transforms as s varies from 0 to $q - 1$. If we pick the a th component of each of these transforms, and use the series constructed as the input for a q -point DLCT, we can obtain $Z^A(aq + b)$, which is called the a th frequency cell. We note that all N sample points are used to obtain the spectrum of the predetermined frequency band. Therefore, the resolution is equivalent to that of direct N -point FLCT. However, the output result is not equivalent to the spectrum of LCT due to supposing $e^{-j\pi2vb/N} = 1$ and $e^{j\pi\alpha'2aqb/(N|\beta|)} = 1$. To use Eq. (20) to approximate LCT, the correction terms must be derived to compensate the effect of $e^{-j\pi2vb/N} = 1$ and $e^{j\pi\alpha'2aqb/(N|\beta|)} = 1$. Thus, we have to select specific types of signals to achieve this goal. Since a chirp signal is a typical non-stationary signal, which widely appears in systems such as communication, radar, sonar, biomedicine, and earthquake exploring systems and also can be used as signal model for many natural phenomena, the correction terms for chirp signal will be discussed in the next subsection.

2. Corrections for ZFLCT of Chirp Signals

Let the specific types of chirp signal be modeled as

$$x(t) = e^{j\pi\mu t^2 + j2\pi f_0 t}, \quad (21)$$

where μ is chirp rate, and f_0 is initial frequency. According to the orthogonality property of LCT basis, Eq. (21) will produce an impulse associating with $\mu = -\gamma$ at $u = f_0/\beta$ in the LCT domain. Therefore, we derive the correction term in case of $\mu = -\gamma$.

We sample the signal $x(t)$ by the interval Δt :

$$x[(sp + v)\Delta t] = e^{j\pi\mu[(sp + v)\Delta t]^2 + j2\pi f_0(sp + v)\Delta t}. \quad (22)$$

We suppose the following parameters:

$$f_0 p \Delta t = a_0 + \delta, b_0 = q\delta,$$

where a_0 is the integer closest to $f_0 p \Delta t$. Thus, Eq. (22) can be rewritten as

$$x[(sp + v)\Delta t] = e^{j\pi\mu(sp\Delta t)^2} e^{j2\pi s\delta} e^{j\pi A \Delta t^2 2spv} e^{j\pi\mu(v\Delta t)^2} e^{j2\pi v(\frac{a_0 + \delta}{p})}. \quad (23)$$

By substituting Eq. (23) into Eq. (19), we have

$$Y_1(s, a_0) = e^{j\pi\mu(sp\Delta t)^2} e^{j2\pi s\delta} e^{j\pi\alpha' \frac{(a_0 q)^2}{N|\beta|}} e^{j\pi \left(\frac{b_0}{q} - \frac{b_0}{N}\right)} \frac{\sin(\pi\delta)}{\sin(\pi \frac{\delta}{p})}. \quad (24)$$

By substituting Eq. (24) into Eq. (20), the ZFLCT is as follows:

$$Z^A(a_0 q + b) = e^{j\pi\alpha' \frac{(a_0 q)^2}{N|\beta|}} e^{j\pi\alpha' \frac{b^2}{N|\beta|}} e^{j\pi \left(b_0 - \frac{b_0}{N} + \frac{b}{q}\right)} \times \frac{\sin(\pi\delta)}{\sin(\pi \frac{\delta}{p})} \frac{\sin(\pi b_0)}{\sin(\pi \frac{b-b_0}{q})}, \quad (25)$$

where $b = 0, 1, 2, \dots, q - 1$.

On the other hand, a direct N -point DLCT can be calculated with the result for the $a_0 q + b$, $b = 0, 1, 2, \dots, q - 1$ component, so from Eq. (18), we have

$$X^A(a_0 q + b) = e^{j\pi\alpha' \frac{(a_0 q)^2}{N|\beta|}} e^{j\pi\alpha' \frac{2aqb}{N|\beta|}} e^{j\pi\alpha' b^2} e^{j\pi \left(b_0 + \frac{b}{N} - \frac{b_0}{q}\right)} \frac{\sin(\pi b_0)}{\sin(\pi \frac{b-b_0}{N})}. \quad (26)$$

By comparing Eqs. (25) and (26), we obtain an amplitude correction as

$$\left| \frac{R_X}{R_Z} \right| = \left| \frac{\sin(\pi \frac{b_0}{N}) \sin[\pi(\frac{b}{q} - \frac{b_0}{q})]}{\sin[\pi(\frac{b}{N} - \frac{b_0}{N})] \sin(\pi \frac{b_0}{q})} \right| \quad (27)$$

and a phase correction as

$$\Delta\theta = \pi b \left(\alpha' \frac{2a_0 q}{N|\beta|} + \frac{1}{N} - \frac{1}{q} \right), \quad (28)$$

where the $|R_X|$ and $|R_Z|$, respectively, are amplitudes of $X^A(a_0 q + b)$ and $Z^A(a_0 q + b)$. The ZFLCT associated with these correction terms is called CZFLCT in this paper. In the next subsection, the implementation and complexity analysis of ZFLCT are investigated in detail.

3. Implementation and Complexity Analysis of ZFLCT

It is interesting to note that Eq. (20) can be further put in a matrix form, as

$$\mathbf{Y}_1^{A''}(a) = \mathbf{U} \mathbf{x}_s(v), \quad s = 0, 1, \dots, q - 1, \quad (29)$$

$$\mathbf{Z}_a^A(b) = \mathbf{V} \mathbf{Y}_1^{A''}(s), \quad b = 0, 1, \dots, q - 1, \quad (30)$$

where $a = 0, 1, \dots, p - 1$,

$$\mathbf{x}_s(v) = [x(sp), e^{j\pi\gamma' \frac{2s}{N|\beta|}} x(sp + 1), \dots, e^{j\pi\gamma' \frac{2s(p-1)}{N|\beta|}} x(sp + p - 1)]^T,$$

$$\mathbf{Y}_1^{A''}(a) = [Y_1(0, s), Y_1(1, s), \dots, Y_1(p - 1, s)]^T,$$

$$\mathbf{Y}_1^{A''}(s) = [Y_1(a, 0), Y_1(a, 1), \dots, Y_1(a, q - 1)]^T,$$

$$\mathbf{Z}_a^A(b) = [z(aq), z(aq + 1), \dots, z(aq + q - 1)]^T.$$

For certain a_0 , we can calculate the spectrum in the frequency band $a_0 q + b$, $b = 0, 1, 2, \dots, q - 1$. Thus, the algorithm is implemented by the following steps:

Step 1: Divide the input sequence $x(n)$ into a $p \times q$ matrix, denoted by $\mathbf{x}_{p \times q} = \{x(sp + v)\}_{v=s=0}^{p-1, q-1}$, where s and v , respectively, are column and row of matrix $\mathbf{x}_{p \times q}$, $s = 0, 1, \dots, q - 1$; $v = 0, 1, \dots, p - 1$.

Step 2: Each element $\{x(sp + v)\}_{v=s=0}^{p-1, q-1}$ of matrix $\mathbf{x}_{p \times q}$, multiplied by the $e^{j\pi\gamma' 2sv/(q|\beta|)}$, is denoted by

$$\mathbf{x}_{p \times q} = \{x(sp + v) e^{j\pi\gamma' 2sv/(q|\beta|)}\}_{v=s=0}^{p-1, q-1}.$$

Step 3: Perform q of FLCTs for all columns of $\mathbf{x}_{p \times q}$ with parameters $\alpha'q, \beta$, and γ'/q , denoted by

$$\mathbf{Y1} = [\mathbf{Y1}_0^{A''}(a), \mathbf{Y1}_1^{A''}(a), \dots, \mathbf{Y1}_{q-1}^{A''}(a)].$$

Step 4: Transpose matrix $\mathbf{Y1}$ and perform a FLCT with parameters α'/p , β , and $\gamma'p$ for the a_0 th column, thereby obtaining the specific interval spectrum $\mathbf{Z}_{a_0}^A(b)$.

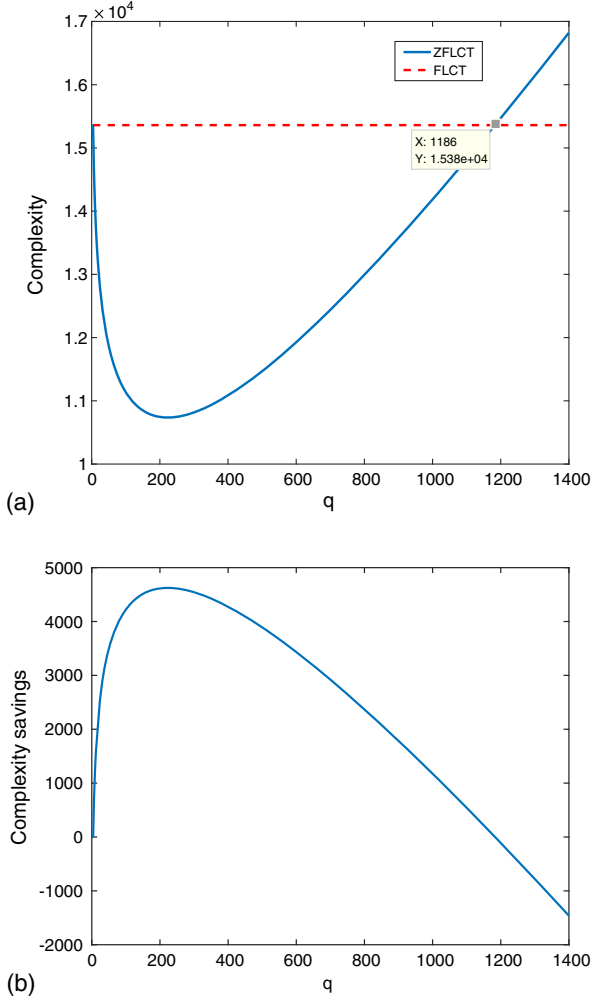


Fig. 2. When $N = 2048$, (a) the number of calculations for FLCT and ZFLCT and (b) computational savings between FLCT and ZFLCT.

According to the above analysis, the ZFLCT performs q of the p -point transform in Eq. (29). Then, $\mathbf{Y1}_a^{A''}(s)$ is used in producing the vector $\mathbf{Z}_a^A(b)$. Here, $\mathbf{Y1}_a^{A''}(s)$ is the a th column of the transpose of matrix $\mathbf{Y1}$, and $\mathbf{Z}_a^A(b)$ contains spectral components in the a th cell with resolution $f/(N|\beta|)$. The total computation is

$$qT_p + T_q + M(N) = 3N + \frac{N}{2} \log_2(p) + 2q + \frac{q}{2} \log_2(q).$$

To obtain the same resolution, if the core storage can handle N sample points, we require for direct N -point FLCT

$$T_N = 2N + \frac{N}{2} \log_2(p) + \frac{N}{2} \log_2(q),$$

thus the calculation savings is given by

$$T_N - (qT_p + T_q + N) = \frac{1}{2}(N - q) \log_2(q) - 2q - N.$$

The savings relates to the length of input data N and factor q . To verify the effect of the two factors on the amount of calculation, we present a simulation. Figures 2(a) and 2(b) show the trends of computation with q for given $N = 2048$. It is interesting to note that the ZFLCT has a lower computation burden than FLCT in the case of the factor $q < 1182$.

For a chirp signal, in order to handle the selected frequency cell $a_0q + b$, $b = 0, 1, 2, \dots, q-1$ by CZFLCT, the result $\mathbf{Z}_{a_0}^A(b)$ needs to be reprocessed using Eqs. (28) and (27) to obtain accurate phase and amplitude. Thus, the number of computations is $3N + N/2 \log_2(p) + 3q + q/2 \log_2(q)$, and the calculation savings is $1/2(N - q) \log_2(q) - N - 3q$. In the case of $2N - N/2 \log_2(N) > 2q + q/2 \log_2(q) - N \log_2(q)$, the CZFLCT has lower computational burden than the SFLCT.

4. SIMULATIONS

To demonstrate the effectiveness of the proposed algorithms, some simulation results are given. Since chirp signals are suitable for the SFLCT, ZFLCT, and CZFLCT, we adopt chirp signals in the following simulation experiments.

The first sets are mono-component chirp signals, as

$$x(t) = B e^{j\pi\mu t^2 + j2\pi f_0 t},$$

Table 1. Comparisons of FLCT, ZFLCT, CZFLCT, and SFLCT for Case1 and Case2

Frequency	FLCT [24] and SFLCT		ZFLCT		CZFLCT	
	Amplitude	Phase	Amplitude	Phase	Amplitude	Phase
<i>case1</i>	$\times 10^3$	$\times 10$	$\times 10^3$	$\times 10$	$\times 10^3$	$\times 10$
1.0000	9.5794	-0.2514	9.5794	-0.2514	9.5794	-0.2514
1.0625	2.3949	0.0225	2.4322	0.1008	2.3949	0.0225
1.1250	1.0644	-0.0203	1.1573	0.1364	1.0644	-0.0203
1.1875	0.6842	-0.0656	0.8435	0.1696	0.6842	-0.0656
1.2500	0.5042	-0.1133	0.7539	0.2003	0.5042	-0.1133
<i>case2</i>	$\times 10^3$	$\times 10$	$\times 10^3$	$\times 10$	$\times 10^3$	$\times 10$
8.1071	0.0041	0.1084	0.0009	-0.0507	0.0041	0.1084
8.1429	0.0054	-0.2331	0.001	-0.0264	0.0054	0.2331
8.1786	0.0082	0.0494	0.0013	-0.0063	0.0082	0.0494
8.2143	0.0163	-0.3006	0.0023	0.0095	0.0163	-0.3006
8.2500	2.0478	-0.0266	0.2826	0.0211	2.0478	-0.0266

Table 2. Comparison of the Number of Computations of FLCT, ZFLCT, CFLCT, and SFLCT for Case3

Method	FLCT	ZFLCT	CZFLCT	SFLCT
Computational Complexity	15,360	14,364	14,372	13,320
Computational Savings	0	996	988	2040

where

$$\text{case1} \quad B = 5, \mu = 1, f_0 = 2.025;$$

$$\text{case2} \quad B = 1, \mu = -5.35, f_0 = 28.876;$$

$$\text{case3} \quad B = 4, \mu = -3.45, f_0 = 40.520.$$

These signals are sampled with sampling interval $\Delta t = 1/256$, and the number of sampling points is $N = 2048$. The factors $p = 256$ and $q = 8$ are applied to algorithms SFLCT, ZFLCT, and CZFLCT. Direct N -point FLCT is also calculated for the sake of comparison.

The LCT parameters $[\alpha, \beta, \gamma]$ of *case1*, *case2*, and *case3* are $[-1, 2, -1]$, $[-5.32, 3.5, -5.35]$, and $[3, 4, 3.45]$, respectively. The peaks of signals in *case1* and *case2* appear near the border of a given frequency cell. The peak of signal *case3* is near the center of a given frequency cell.

The results are tabulated in Table 1 for the chirp signals *case1* and *case2*. Figure 3(a) represents the amplitude in the relevant frequency cell for *case3*. These results suggest that the amplitude components obtained using ZFLCT are different from the direct N -point FLCT, but they have the same spectral shape and peak position. Thus, the ZFLCT can be applied effectively to chirp signal detection. However, the results of CZFLCT are almost coincident with the direct FLCT in the relevant frequency cell. The amplitude errors between FLCT and CZFLCT, SFLCT are shown in Fig. 3(b). “CZFLCT-FLCT” and “SFLCT-FLCT”, respectively, denote amplitude error between CZFLCT and FLCT, SFLCT and FLCT in Fig. 3. The phase error is shown in Fig. 3(c). It is clear that the SFLCT and CZFLCT are nearly equivalent to FLCT. The number of computations is shown in Table 2, and the calculation savings is obvious. Therefore, when we are interested in only certain regions of frequency, the CZFLCT and SFLCT are more effective than FLCT.

The remaining two signals are multi-component chirp signal as follows:

$$x(t) = Be^{j\pi\mu t^2 + j2\pi f_0 t} + Ce^{j\pi\mu t^2 + j2\pi f_1 t},$$

where, for signal

$$\text{case4} \quad B = 1, C = 1, \mu = 2.82, f_0 = 1.1 \text{ Hz}, f_1 = 2.2 \text{ Hz};$$

$$\text{case5} \quad B = 10, C = 15, \mu = 3.55, f_0 = 7.876 \text{ Hz}, f_1 = 7.86 \text{ Hz}.$$

Case4 contains two initial frequencies that are lying in the adjacent two frequency cells, and the two signals have the same amplitude. *Case5* contains two very close frequencies with different amplitudes.

We take $N = 40960$, $p = 2560$, $q = 16$, the sampling interval $T = 1/p$, and LCT parameters $[2.6, 3, -2.82]$. Only amplitude and phase complements near the expected frequencies are presented. The amplitude and phase of *case4* in the relevant cells are given in Fig. 4 and Table 3, respectively.

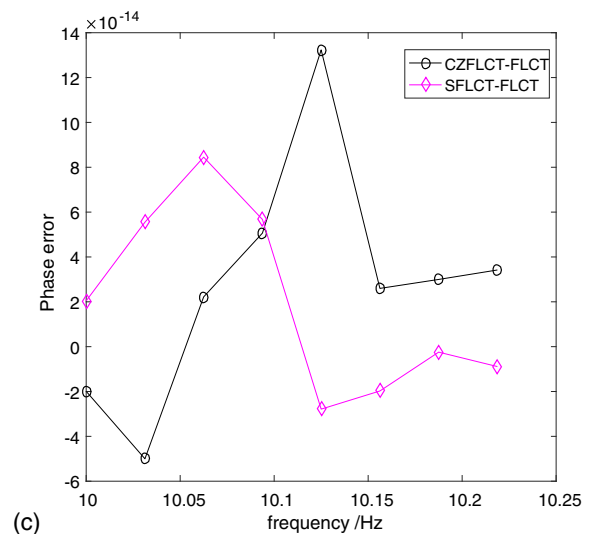
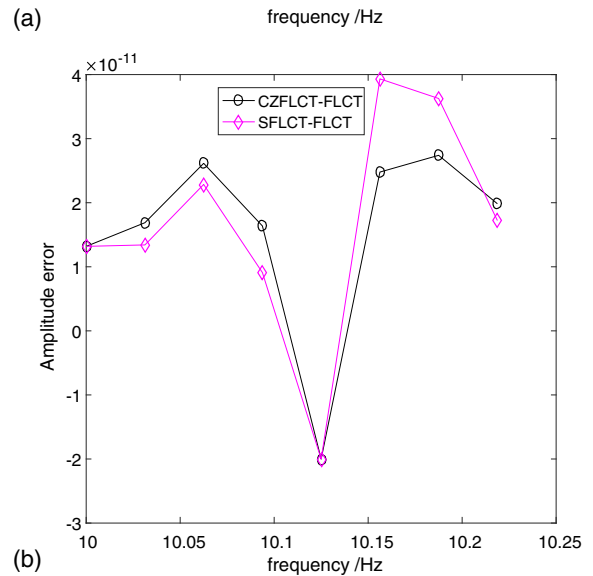
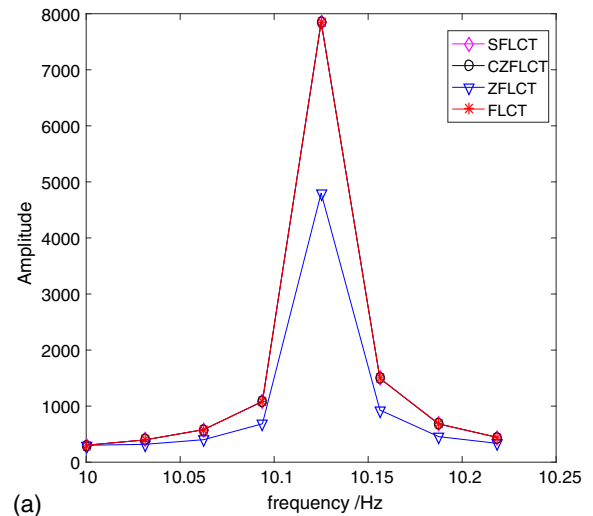
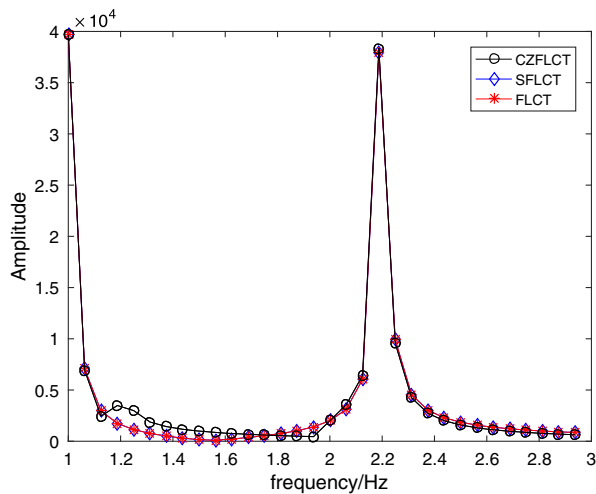


Fig. 3. Result of *case3* in local frequency interval $[10.00, 10.219]$. (a) Amplitudes of FLCT, ZFLCT, CZFLCT and SFLCT. (b) Amplitude error of CZFLCT, SFLCT and FLCT. (c) Phase error of CZFLCT, SFLCT and FLCT.

Table 3. Phases of FLCT, CZFLCT, and SFLCT for Case4

Frequency	FLCT [24]	CZFLCT	SFLCT
	Phase	Phase	Phase
<i>case4</i>			
1.0000	1.4115	1.4115	1.4115
1.0625	-1.6217	-1.6243	-1.6217
1.1250	-1.5087	-1.5401	-1.5087
1.2500	-1.2755	-1.1641	-1.3925
2.000	-2.0003	-2.0003	-2.0003
2.0625	-1.7795	-1.7970	-1.7795
2.1250	-1.5495	-1.5574	-1.5495
2.1875	-1.3109	-1.3121	-1.3109

**Fig. 4.** Amplitudes of FLCT, CZFLCT, and SFLCT in frequency interval [1, 2.9375] for *case4*.

For chirp signal *case5*, we take $N = 20480$, $p = 80$, $q = 256$ associated with LCT parameters $[1, 2, -5.35]$. We note the initial frequencies in LCT domain are resolved by ZFLCT, SFLCT, and CZFLCT in Fig. 5. The number of computations for obtaining the local spectrum is shown in Table 4.

Therefore, all simulation results show that SFLCT and CZFLCT have the same performance as FLCT in terms of accuracy. However, the proposed algorithms can implement the selectivity of the spectrum output, and the calculation is reduced by concentrating the analysis on selected frequency bands. When each frequency cell is calculated, all sampling points are used, thus, the resolution is $1/(N|\beta|\Delta t)$. Therefore, SFLCT and ZFLCT can improve the resolution for limited analysis equipment. In addition, the correction term of ZFLCT, which is to obtain accurate amplitude and phase of the signal, is only valid for chirp signals. The ZFLCT can provide the same peak position for the chirp signal as FLCT, and it can be used to detect chirp signals. From Table 4, we conclude that ZFLCT has a lower computational burden than SFLCT.

5. CONCLUSIONS

In this paper, we develop the SFLCT and ZFLCT. The advantages of the two proposed algorithms are that they can handle very large input data with very limited memory space, and presents a lower complexity burden than FLCT by

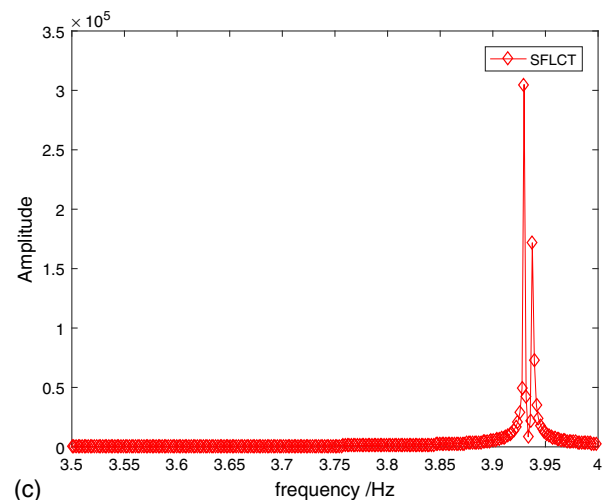
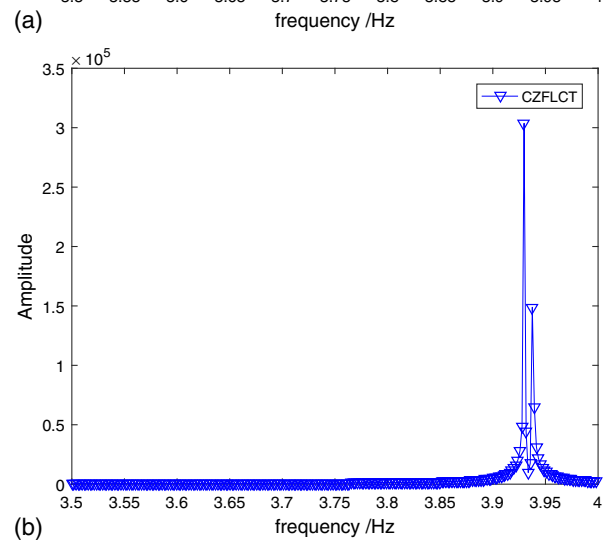
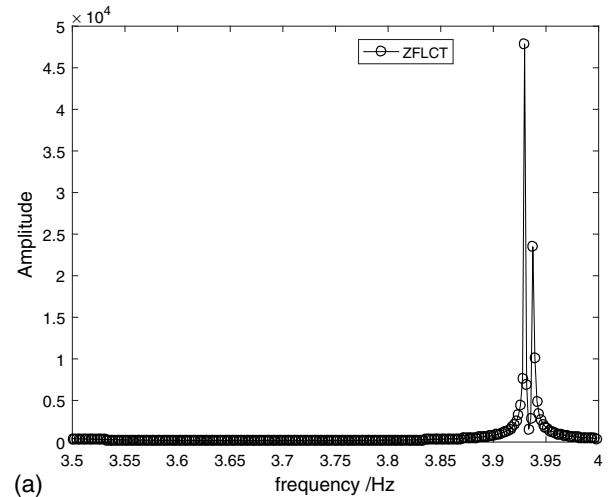
**Fig. 5.** For *case5*, spectrum calculated in local interval [3.5, 4] by different algorithms: (a) ZFLCT, (b) CZFLCT, and (c) SFLCT.

Table 4. Comparison of the Complexity of FLCT, ZFLCT, CZFLCT, and SFLCT for Case5

Method	FLCT	ZFLCT	CZFLCT	SFLCT
Calculational Complexity	1.8762×10^5	1.2771×10^5	1.2797×10^5	1.84576×10^5
Computational Savings	0	5.9904×10^4	5.9648×10^4	3.044×10^3

concentrating the analysis on selected frequency bands. Both of them have respective excellent features. The SFLCT has fewer restrictions on the processed signals. Nevertheless, the ZFLCT has a lower computational burden than SFLCT in some cases. In addition, in the aspect of program implementation, the ZFLCT is easier than the SFLCT. Their performance with regard to resolution, computational complexity, and precision is compared with that of the FLCT. The simulations also show that the proposed algorithms are equivalent to FLCT in their respective appropriate conditions. The advantages of proposed algorithms can be summarized as follows:

- Only a small size of the DLCT is used to assist the implementation. Hence, the proposed algorithms are able to handle very large input data with very limited memory.
- They are able to calculate a portion of the frequency spectrum of interest instead of the whole spectrum as the FLCT does.
- An ideal frequency resolution can be reached because all the input data are used.
- The proposed algorithms have a lower number of computations than the direct N -point FLCT for calculating the LCT spectrum in a particular band.

As a conclusion, we recommend the SFLCT and ZFLCT as basic means to calculate the DLCT of a large amount of input data in a specific frequency region of interest. We can select different implementation methods according to the type of signal. More applications of the proposed SFLCT and ZFLCT still need to be explored.

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